SOLUÇÕES FICHA 5

C. TÓPICOS ADICIONAIS

C.4. Introdução às Transformadas de Laplace

5.1.

a) $\frac{1}{s^2}(1-e^{-s}), s>0$

b) $\frac{1}{s^2+1}(e^{-\pi}+1), s>0$

c) $\frac{e^{-5}}{s+2}, s>-2$

d) $\frac{2}{(s+2)^3}, s>-2$

e) $\frac{1}{(s+2)^2+1}, s>-1$

f) $\frac{s^2-1}{(s+1)^2}, s>0$

5.2.

a) $\frac{2\cdot 4!}{s^5}$

b) $\frac{5!}{s^6}$

c) $\frac{4}{s^2}-\frac{10}{s}$

d) $\frac{7}{s^2}+\frac{3}{s}$

e) $\frac{2}{s^3}+\frac{6}{s^2}-\frac{3}{s}$

f) $\frac{2}{s^3}-\frac{1}{s+9}+\frac{5}{s}$

g) $\frac{s}{s^2+25}+\frac{10}{s^2+4}$

h) $\frac{k}{s^2-k^2}+\frac{s}{s^2-k^2}$

i) $\frac{1}{2(s-2)}-\frac{1}{2s}$

j) $\frac{1}{2(s+2)}+\frac{1}{2s}$

k) $\frac{2}{s^2+16}$

l) $\frac{1}{2s}+\frac{s}{2(s^2+4)}$

5.3.

a) $\frac{1}{3}\sin(3t)$

b) $\frac{5}{3}e^{\frac{1}{3}t}$

c) $3t^2$

d) $\frac{1}{2}t^3$

e) $\frac{3}{\sqrt{7}}\sinh(\sqrt{7}t)$

f) $\frac{1}{12}e^{3t}t^4$

g) $7\cos(2t)$

h) $7\cosh(4t)$

i) $e^{2t}\sin(3t)$

j) $2e^{-t}\cos(3t)$

k) $4e^{-2t}\cosh(3t)+\frac{5}{3}e^{2t}\sinh(3t)$

l) $(t+2)e^{-t}-e^{-2t}$

C.5. Introdução às Séries de Fourier

5.4.

$$f(x) = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\sin[(2n-1)x]}{(2n-1)}$$

5.5.

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty}\frac{4(-1)^n}{n^2}\cos(nx)$$

5.6.

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{2}{(2n-1)\pi}\sin[(2n-1)\pi x]$$

5.7.

$$g(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{2}{(2n-1)\pi}\sin[(2n-1)\pi x]$$

5.8.

$$f(x) = \frac{1}{4} + \sum_{n=1}^{\infty}\left[\frac{(-1)^n - 1}{n^2\pi^2}\cos(n\pi x) + \frac{(-1)^{n+1}}{n\pi}\sin(n\pi x)\right]$$

5.9.

$$f(x) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty}\left[\frac{2(-1)^n}{n^2}\cos(nx) + \left(\frac{\pi(-1)^{n+1}}{n} + \frac{\pi 2}{\pi n^3}((-1)^n - 1)\right)\sin(nx)\right]$$